Information Geometry of the Quantum-Classical Transition: From Photosynthetic Excitons to Neural Oscillations

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Abstract

Decoherence drives the quantum-to-classical transition, but its information-geometric structure—the precise dimensional and metric changes it imposes on quantum state space—has not been quantified in the Bures/Fisher–Rao framework. We show that decoherence is a dimensional collapse: the dephasing channel maps a (d^2-1)-dimensional quantum manifold onto the $(d-1)$-dimensional classical probability simplex, annihilating the d^2-d coherence directions while preserving the $d-1$ population dimensions. At diagonal states, the classical and quantum tangent subspaces are Bures-orthogonal. We compute the exact Bures principal angle θ_B between these subspaces at every full-rank qubit state, obtaining $\cos^2\theta_B = r_z^2 r_\perp^2/[(1-r_\perp^2)(1-r_z^2)]$, where r_z is the population imbalance and $r_\perp$ the coherence magnitude. For $d \geq 3$, we prove that Bures-orthogonality holds if and only if ρ is diagonal—any coherence breaks the clean classical-quantum split. We apply this framework to the Fenna–Matthews–Olson photosynthetic complex ($d = 7$), where decoherence collapses 48 quantum dimensions to 6 classical di-

mensions. Using a Haken–Strobl model with reaction-centre trapping, we compute transport efficiency and the six Bures principal angles from the same sink-inclusive dynamics, with the geometry extracted from the renormalized conditional state. At the ENAQT-optimal dephasing rate ($\gamma_{\mathrm{opt}} \approx 100$–$150\,\mathrm{cm}^{-1}$), all principal angles exceed 87°: biologically optimal transport occurs in a regime that is geometrically close to classical, with strong quantum-classical mixing confined to weaker dephasing ($\gamma \lesssim 10\,\mathrm{cm}^{-1}$). The same near-classical geometric signature is reproduced in the structurally distinct PE545 cryptophyte antenna ($d = 8$, collapse $63 \to 7$), suggesting this property is generic to environment-assisted transport. Extending to a cortical gamma-oscillation assembly ($d = 10$, collapse $99 \to 9$), we find that biological dephasing places neural systems ∼12 orders of magnitude past the quantum-classical boundary: geometrically indistinguishable from the classical simplex at oscillation-frequency scales, though the framework quantifies the detection threshold for any residual quantum structure.

1 Introduction

The quantum-to-classical transition mediated by decoherence [1, 2, 3, 4] is not merely a theoretical curiosity in biological systems—it is functionally relevant. Photosynthetic complexes exploit quantum coherence during energy transfer on femtosecond timescales [5], and optimal transport efficiency occurs at intermediate dephasing rates, a phenomenon termed environment-assisted quantum transport (ENAQT) [6, 7, 8]. The quantum biology literature has established that partial decoherence can enhance biological function [9], but the geometric structure of this transition—how many degrees of freedom are destroyed, how cleanly classical information separates from quantum information, and what the dimensional architecture of intermediate regimes looks like—has not been quantified.

Information geometry provides the natural mathematical language for these questions. The quantum Fisher information matrix (QFIM) [10, 11] measures the dimensionality of distinguishable quantum states, and the Bures metric it induces [12, 13, 14] is the unique

minimal monotone Riemannian metric on quantum state space. The classical Fisher–Rao metric [15, 16, 17] plays the analogous role for probability distributions. Despite the maturity of both decoherence theory and information geometry, these frameworks have not been combined to characterize the dimensional structure of the quantum-to-classical transition.

In this paper, we show that decoherence is a *dimensional collapse* in the information-geometric sense. A d-dimensional quantum system lives on a (d^2-1)-dimensional state manifold equipped with the Bures metric. Decoherence projects this manifold onto the $(d-1)$-dimensional classical probability simplex equipped with the Fisher–Rao metric. Specifically:

(Collapse.) The decoherence channel reduces the QFIM rank from d^2-1 to at most $d-1$ (Theorem 3.2), and the Bures metric restricts to the Fisher–Rao metric on Δ_{d-1} (Theorem 3.3). At full-rank diagonal states, the differential of the dephasing map is the Bures-orthogonal projector onto the classical tangent subspace (Theorem 3.4).

metry of the split.) We compute the exact Bures principal angle θ_B between the classical and quantum tangent subspaces at every full-rank qubit state (Theorem 3.5). For $d \geq 3$, we prove that Bures-orthogonality holds if and only if ρ is diagonal (Theorem 3.8), and a perturbative formula quantifies the departure (Theorem 3.10).

(Application.) We apply the framework to three biological systems spanning 12 orders of magnitude in dephasing: two photosynthetic complexes—the Fenna–Matthews–Olson (FMO) complex ($d = 7$, collapse $48 \to 6$) and the PE545 cryptophyte antenna ($d = 8$, collapse $63 \to 7$)—and a cortical gamma-oscillation assembly ($d = 10$, collapse $99 \to 9$). At the ENAQT-optimal dephasing rate both photosynthetic complexes are geometrically nearly classical (all angles $> 87°$ for FMO, $> 89°$ for PE545). The neural system sits $\sim$12 orders of magnitude past the quantum-classical boundary, geometrically indistinguishable from the classical simplex at oscillation-frequency scales.

The paper is organized as follows. Section 2 establishes the information-geometric

framework. Section 3 contains the core mathematical results. Section 4 derives the thermodynamic cost of dimensional collapse. Section 5 applies the framework to FMO and PE545. Section 6 extends to neural oscillations. Section 7 discusses implications, limitations, and connections to classical biological coherence.

2 Information-Geometric Framework

We establish the mathematical framework that makes "dimensional collapse" precise.

Definition 2.1 (Quantum state manifold)**.** The space of $d \times d$ density matrices,

$$\mathcal{M}_Q = \Big\{\rho \in \mathbb{C}^{d\times d} : \rho \geq 0,\ \mathrm{Tr}\,\rho = 1\Big\}, \tag{1}$$

is a real manifold of dimension $d^2 - 1$: a $d \times d$ Hermitian matrix has d^2 real parameters, the trace constraint removes one, leaving $d - 1$ populations and $d(d-1)$ coherences.

Definition 2.2 (Quantum Fisher information matrix)**.** Let $\rho(\theta)$ be a smooth parametric family of density matrices with $\theta \in \mathbb{R}^p$. The QFIM is the $p \times p$ matrix with entries

$$[F_Q]_{ij} = \mathrm{Tr}\Big[\rho\,\frac{L_iL_j + L_jL_i}{2}\Big], \tag{2}$$

where L_i is the symmetric logarithmic derivative (SLD) defined by $\partial_i\rho = (\rho L_i + L_i\rho)/2$. For full-rank ρ with eigenbasis $\rho = \sum_k p_k|k\rangle\langle k|$, the matrix elements are $(L_i)_{mn} = 2(\partial_i\rho)_{mn}/(p_m + p_n)$.

The QFIM defines the Bures–Helstrom metric $ds^2_{\mathrm{Bures}} = \frac{1}{4}\sum_{ij}[F_Q]_{ij}\,d\theta^i\,d\theta^j$, the unique minimal monotone Riemannian metric on quantum state space [14, 11]. By Petz's monotonicity theorem, for any CPTP map Λ:

$$F_Q\Big(\Lambda[\rho(\theta)]\Big) \leq F_Q\Big(\rho(\theta)\Big) \tag{3}$$

in the Löwner order—quantum channels cannot increase statistical distinguishability.

Definition 2.3 (Effective quantum dimensionality)**.** Given a QFIM F_Q with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$, the effective quantum dimensionality is the participation ratio

$$D_{\text{eff}}^Q = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}, \tag{4}$$

ranging from 1 (single dominant direction) to p (all directions equally estimable). D_{eff}^Q is a coordinate-dependent diagnostic: the rank of F_Q is invariant, but the participation ratio depends on the parameterization. The main theorems of this paper depend only on rank, kernel structure, and metric properties.

3 Decoherence as Dimensional Projection

We now show that decoherence acts as a geometric projection, annihilating the $d^2 - d$ off-diagonal directions and collapsing quantum state space onto the $(d-1)$-dimensional classical probability simplex.

Definition 3.1 (Decoherence channel)**.** Fix an orthonormal pointer basis $\{|k\rangle\}_{k=1}^d$. The complete dephasing channel is

$$\Lambda_{\text{dec}}(\rho) = \sum_{k=1}^{d} |k\rangle\langle k|\, \rho\, |k\rangle\langle k|, \tag{5}$$

setting all off-diagonal elements to zero: $[\Lambda_{\text{dec}}(\rho)]_{kl} = \delta_{kl}\, \langle k|\rho|k\rangle$.

Theorem 3.2 (Dimensional collapse)**.** *Let $\mathcal{M}_Q$ be the quantum state manifold of a d-dimensional system* ($\dim \mathcal{M}_Q = d^2 - 1$)*. The decoherence channel Λ_{dec} maps $\mathcal{M}_Q$ onto the classical probability simplex Δ_{d-1}* ($\dim \Delta_{d-1} = d - 1$)*. Specifically:*

(i) *The image $\Lambda_{\text{dec}}(\mathcal{M}_Q) = \Delta_{d-1}$.*

(ii) *The $d^2 - d$ off-diagonal parameters become unestimable: for any smooth parameterization $\theta = (\theta_{\text{diag}}, \theta_{\text{off}})$, the QFIM of $\Lambda_{\text{dec}}[\rho(\theta)]$ satisfies*

$$[F_Q(\Lambda_{\text{dec}}[\rho])]_{\alpha\beta} = 0 \quad \textit{whenever } \alpha \in \theta_{\text{off}} \textit{ or } \beta \in \theta_{\text{off}}. \tag{6}$$

In particular, $\operatorname{rank} F_Q(\Lambda_{\mathrm{dec}}[\rho]) \leq d-1$.

(iii) *The dimensional collapse ratio is*

$$\frac{\dim \Delta_{d-1}}{\dim \mathcal{M}_Q} = \frac{d-1}{d^2-1} = \frac{1}{d+1}, \tag{7}$$

which vanishes as $d \to \infty$.

Proof. Part (i): Λ_{dec} sends ρ to $\mathrm{diag}(\langle k|\rho|k\rangle)$, a point in Δ_{d-1}; every point in Δ_{d-1} is the image of some diagonal state. Part (iii) is arithmetic.

For Part (ii): the decohered state depends only on θ_{diag}, so $\partial\rho_{\mathrm{dec}}/\partial\theta^{\alpha}_{\mathrm{off}} = 0$, giving $L_\alpha = 0$ and hence $[F_Q]_{\alpha\beta} = 0$ whenever α or β is off-diagonal. The surviving diagonal block reduces to the classical Fisher information:

$$[F_Q]^{\mathrm{diag}}_{ij} = \sum_{k=1}^{d} \frac{1}{p_k} \frac{\partial p_k}{\partial \theta^i_{\mathrm{diag}}} \frac{\partial p_k}{\partial \theta^j_{\mathrm{diag}}} = [F_{\mathrm{cl}}]_{ij}, \tag{8}$$

which has rank at most $d-1$. □

Theorem 3.3 (Metric restriction)**.** *Under* Λ_{dec}*, the Bures metric on* $\mathcal{M}_Q$ *restricts to the Fisher–Rao metric on* Δ_{d-1}*:*

$$ds^2_{\mathrm{Bures}}\Big|_{\Delta_{d-1}} = \frac{1}{4}\sum_{k=1}^{d} \frac{dp_k^2}{p_k} = \frac{1}{4}\, ds^2_{\text{Fisher-Rao}}. \tag{9}$$

Proof. For diagonal states, the SLD is diagonal: $\langle k|L_i|k\rangle = (\partial_i p_k)/p_k$. The Bures line element becomes $\frac{1}{4}\sum_k dp_k^2/p_k$, the Fisher–Rao metric [11]. □

Theorem 3.4 (Orthogonal decomposition)**.** *Let* $\rho = \mathrm{diag}(p_1, \ldots, p_d)$ *be a full-rank diagonal state. The tangent space admits an orthogonal decomposition under the Bures metric:*

$$T_\rho \mathcal{M}_Q = T_\rho^{\mathrm{diag}} \oplus T_\rho^{\mathrm{off}}, \tag{10}$$

where T_ρ^{diag} *is the* $(d-1)$*-dimensional traceless diagonal subspace and* T_ρ^{off} *is the* (d^2-d)*-dimensional off-diagonal Hermitian subspace. The differential* $d\Lambda_{\mathrm{dec}}|_\rho$ *acts as the identity on* T_ρ^{diag} *and as zero on* T_ρ^{off}*, so it is the* g_B*-orthogonal projector onto* T_ρ^{diag}*.*

Proof. At a diagonal state, any traceless Hermitian X decomposes as $X = X_D + X_O$ (diagonal + off-diagonal). The SLD for a diagonal perturbation X_D is diagonal: $(L_{X_D})_{kk} = x_k/p_k$. The SLD for an off-diagonal perturbation X_O is off-diagonal: $(L_{X_O})_{mn} = 2z_{mn}/(p_m + p_n)$ for $m \neq n$. The anticommutator $\{L_{X_D}, L_{X_O}\}$ is off-diagonal (product of diagonal and off-diagonal matrices), so $\rho\{L_{X_D}, L_{X_O}\}$ is off-diagonal with zero trace:

$$g_B(X_D, X_O) = \frac{1}{4}\,\mathrm{Tr}\left[\rho\,\frac{\{L_{X_D}, L_{X_O}\}}{2}\right] = 0.$$

The dephasing channel preserves diagonal and annihilates off-diagonal perturbations, completing the identification as orthogonal projector. □

3.1 The classical–quantum split away from the simplex

Theorem 3.4 establishes orthogonality at diagonal states. What happens at states carrying coherence? For qubits, a single principal angle characterizes the full relationship.

Theorem 3.5 (Bures separation angle)**.** *Let $\rho = (I + \mathbf{r}\cdot\boldsymbol{\sigma})/2$ be a full-rank qubit state ($|\mathbf{r}| < 1$) with coherence magnitude $r_\perp = \sqrt{r_x^2 + r_y^2}$ and population imbalance r_z. The principal angle θ_B between T_ρ^{diag} and T_ρ^{off} under the Bures metric satisfies*

$$\cos^2\theta_B = \frac{r_z^2\, r_\perp^2}{(1 - r_\perp^2)(1 - r_z^2)}. \tag{11}$$

Equivalently,

$$\sin^2\theta_B = \frac{1 - |\mathbf{r}|^2}{(1 - r_\perp^2)(1 - r_z^2)}. \tag{12}$$

Proof. The Bures metric on qubit state space in Bloch coordinates is $g_{ij}^B = \frac{1}{4}[\delta_{ij} + r_i r_j/(1 - |\mathbf{r}|^2)]$. The principal angle is found by maximizing $g_B(\hat{e}_z, v)/(\|\hat{e}_z\| \cdot \|v\|)$ over $v \in T_\rho^{\mathrm{off}}$.

The maximizer is $v_\perp = (r_x, r_y, 0)/r_\perp$. Computing:

$$g_B(\hat{e}_z, v_\perp) = \frac{r_z r_\perp}{4(1-|\mathbf{r}|^2)},$$
$$g_B(\hat{e}_z, \hat{e}_z) = \frac{1-r_\perp^2}{4(1-|\mathbf{r}|^2)},$$
$$g_B(v_\perp, v_\perp) = \frac{1-r_z^2}{4(1-|\mathbf{r}|^2)}.$$

The squared cosine follows; the sine formula uses $(1-r_\perp^2)(1-r_z^2) - r_z^2 r_\perp^2 = 1-|\mathbf{r}|^2$. □

Corollary 3.6 (Structure of the classical–quantum split)**.**

(i) $\theta_B = \pi/2$ *if and only if* $r_\perp = 0$ *or* $r_z = 0$*: subspaces are Bures-orthogonal at all diagonal and all equatorial states.*

(ii) $\theta_B < \pi/2$ *if and only if both* $r_\perp \neq 0$ *and* $r_z \neq 0$*.*

(iii) *Dephasing monotonically increases* θ_B *toward* $\pi/2$*.*

Proof. Parts (i)–(ii) are immediate from (11). For (iii): under dephasing, $r_\perp \mapsto (1-2\gamma)r_\perp$ while r_z is preserved. The numerator of $\cos^2\theta_B$ decreases monotonically and the denominator increases, so θ_B increases. □

Remark 3.7 (Estimation-theoretic interpretation)*.* For a full-rank qubit, eliminating coherence parameters via the Schur complement gives effective Fisher information $[F_Q]_{zz}^{\text{eff}} = [F_Q]_{zz} \sin^2\theta_B = 1/(1-r_z^2)$ [18, 19]: the Bures separation angle determines the fraction of population Fisher information surviving marginalization over coherence.

3.2 Generalization to qudits

For d-dimensional systems, there are $d-1$ principal angles between T_ρ^{diag} and T_ρ^{off}. Theorem 3.4 shows these are all $\pi/2$ at diagonal states. For $d \geq 3$, the converse holds exactly:

Theorem 3.8 (Global orthogonality criterion for $d \geq 3$)**.** *For* $d \geq 3$*, a full-rank state* ρ *achieves the Bures-orthogonal decomposition* $T_\rho^{\text{diag}} \perp_{g_B} T_\rho^{\text{off}}$ *if and only if* ρ *is diagonal in the pointer basis.*

Proof. The reverse direction is Theorem 3.4. For the forward direction, assume orthogonality holds. Define the Jordan product map $\mathcal{J}_\rho(L) = (\rho L + L\rho)/2$, which is invertible on Hermitian matrices when ρ is full-rank. The SLD for tangent vector X is $L_X = \mathcal{J}_\rho^{-1}(X)$.

For any traceless diagonal $X_D = \mathrm{diag}(x_1, \ldots, x_d)$ and off-diagonal Hermitian X_O, the SLD metric identity gives

$$g_B(X_D, X_O) = \tfrac{1}{2} \sum_k x_k \, (L_{X_O})_{kk}. \tag{13}$$

Since this vanishes for all traceless x, orthogonality requires $(L_{X_O})_{kk}$ to be constant across k—the diagonal of L_{X_O} is flat. Hence $L_{X_O} = cI + Z$ for some $c \in \mathbb{R}$ and Z off-diagonal.

Let $\mathcal{O}$ denote the $d(d-1)$-dimensional space of off-diagonal Hermitian matrices and $\mathcal{W} = \mathbb{R}I \oplus \mathcal{O}$. The flat-diagonal condition means $S := \mathcal{J}_\rho^{-1}(\mathcal{O}) \subset \mathcal{W}$. Consider the projection $\pi \colon \mathcal{W} \to \mathcal{O}$ defined by $cI + Z \mapsto Z$. This is injective on S: if $cI \in S$ then $\mathcal{J}_\rho(cI) = c\rho \in \mathcal{O}$, but ρ has positive diagonal entries, forcing $c = 0$. Since $\dim S = \dim \mathcal{O} = d(d-1)$, the restricted projection $\pi|_S$ is an isomorphism.

Therefore, for every $Z \in \mathcal{O}$ there exists a unique $c(Z) \in \mathbb{R}$ with $c(Z)I + Z \in S$, meaning $\mathcal{J}_\rho(c(Z)I + Z) \in \mathcal{O}$. Taking the diagonal:

$$\mathrm{diag}(\rho Z + Z\rho) = -2c(Z)\, p, \qquad p = (p_1, \ldots, p_d) = \mathrm{diag}(\rho). \tag{14}$$

Now substitute the basis elements. For $R_{kl} = |k\rangle\langle l| + |l\rangle\langle k|$:

$$\mathrm{diag}(\rho R_{kl} + R_{kl}\rho) = 2\,\mathrm{Re}(\rho_{kl})\,(e_k + e_l), \tag{15}$$

and for $I_{kl} = i|k\rangle\langle l| - i|l\rangle\langle k|$:

$$\mathrm{diag}(\rho I_{kl} + I_{kl}\rho) = 2\,\mathrm{Im}(\rho_{kl})\,(e_k + e_l). \tag{16}$$

If $\rho_{kl} \neq 0$, at least one of these is nonzero, so $e_k + e_l \in \mathrm{span}\{p\}$. But for $d \geq 3$, the vector $e_k + e_l$ has $d - 2 \geq 1$ zero entries while p has all entries strictly positive, so $e_k + e_l$ cannot be proportional to p. Contradiction; hence $\rho_{kl} = 0$ for all $k \neq l$. □

Remark 3.9 (The qubit exception). For $d = 2$, the vector $e_1 + e_2 = (1, 1)$ is proportional to $p = (p_1, p_2)$ if and only if $p_1 = p_2$, recovering the qubit result: equatorial states ($r_z = 0$) achieve Bures-orthogonality despite carrying coherence (Corollary 3.6(i)). The proof above shows that $d \geq 3$ eliminates this loophole.

Theorem 3.8 settles the qualitative question: for $d \geq 3$, any coherence breaks orthogonality. We now quantify *how much* the subspaces mix.

Theorem 3.10 (Perturbative cross-Gram near the simplex)**.** *Let $\rho = \mathrm{diag}(p) + C$ be a full-rank d-dimensional state with off-diagonal part C.*

(i) *Near the simplex ($\|C\|$ small), the cross-Gram matrix has entries*

$$g_B(D_a, R_{kl}) = -\frac{\mathrm{Re}[C_{kl}]}{2(p_k + p_l)} \left(\frac{x_a^k}{p_k} + \frac{x_a^l}{p_l} \right) + O(\|C\|^2), \tag{17}$$

where $D_a = \mathrm{diag}(x_a^1, \ldots, x_a^d)$ is a traceless diagonal tangent vector and $R_{kl} = |k\rangle\langle l| + |l\rangle\langle k|$.

(ii) *Under partial dephasing $C_{kl} \mapsto (1-2\gamma)C_{kl}$, the cross-Gram entries scale as $(1-2\gamma)$ at leading order, and all principal angles converge to $\pi/2$ as $\gamma \to 1/2$.*

Proof. Part (i): Expand $L_a = L_a^{(0)} + L_a^{(1)} + O(\|C\|^2)$. At zeroth order, $(L_a^{(0)})_{kk} = x_a^k/p_k$ with zero off-diagonal entries. At first order, the Lyapunov equation gives $(L_a^{(1)})_{mn} = -C_{mn}(x_a^m/p_m + x_a^n/p_n)/(p_m + p_n)$ for $m \neq n$. For R_{kl}, the first-order diagonal corrections are $(L_R^{(1)})_{kk} = -2\mathrm{Re}[C_{kl}]/[p_k(p_k + p_l)]$. Summing the three first-order contributions to $g_B(D_a, R_{kl})$ yields (17).

Part (ii): Under $C \mapsto (1-2\gamma)C$, equation (17) scales as $(1-2\gamma)$ and vanishes as $\gamma \to 1/2$. □

Figure 1 illustrates both results: Theorem 3.8 guarantees no non-diagonal state achieves orthogonality, while the distribution of minimum principal angles shows the typical scale of the departure.

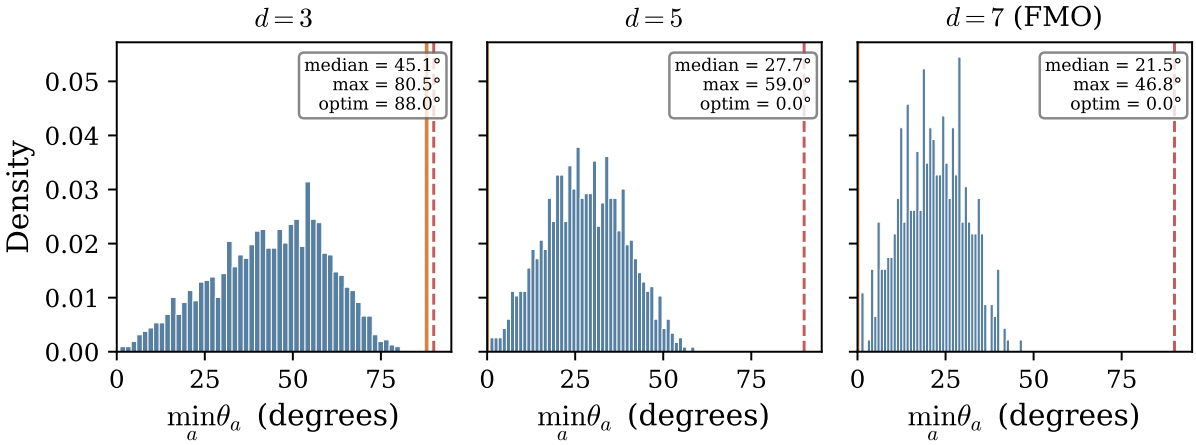

Figure 1: **Numerical illustration of Theorem 3.8.** For each dimension $d \in \{3, 5, 7\}$, random full-rank density matrices are drawn from the Hilbert–Schmidt measure (blue histogram), and the minimum principal angle $\min_a \theta_a$ between the diagonal and off-diagonal tangent subspaces is computed via the Bures metric. The dashed red line marks 90° (perfect orthogonality, achieved only at diagonal states). No non-diagonal state approaches orthogonality; the distribution shifts further from 90° as d increases. The orange line marks the best angle found by adversarial optimization. The $d = 7$ panel corresponds to the FMO Hilbert space dimension.

4 Thermodynamic Cost of Decoherence

Proposition 4.1 (Free energy of coherence)**.** *Let ρ be a quantum state with system Hamiltonian H. The free energy change under dephasing is*

$$F(\rho) - F(\Lambda_{\text{dec}}[\rho]) = \text{Tr}\Big[(\rho - \Lambda_{\text{dec}}[\rho])H\Big] + k_B T\, C_r(\rho), \tag{18}$$

where $F(\rho) = \text{Tr}[\rho H] - k_B T S(\rho)$ is the non-equilibrium free energy and $C_r(\rho) = S(\Lambda_{\text{dec}}[\rho]) - S(\rho) \geq 0$ is the relative entropy of coherence [20].

Proof. Direct rearrangement of $F(\rho) - F(\Lambda_{\text{dec}}[\rho])$. Non-negativity of C_r follows from unitality of Λ_{dec} and Schur-concavity of von Neumann entropy. □

When H is diagonal in the dephasing basis, the energy term vanishes and

$$F(\rho) - F(\Lambda_{\text{dec}}[\rho]) = k_B T\, C_r(\rho) \geq 0. \tag{19}$$

The free energy geometrically decomposes along the orthogonal subspaces of Theorem 3.4:

the classical component on T_ρ^{diag} and the coherent component on T_ρ^{off}.

For a maximally coherent d-dimensional pure state, $C_r = \ln d$, and the cost per collapsed dimension is $k_B T \ln d/(d^2 - d) \approx k_B T \ln d/d^2$, which vanishes with system size—consistent with the extreme rapidity of macroscopic decoherence [3].

5 Application to Photosynthetic Exciton Transport

5.1 The FMO complex

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria (*Chlorobaculum tepidum*) is a trimeric pigment-protein complex that channels excitation energy from the chlorosome antenna to the reaction center. Each monomer contains seven bacteriochlorophyll *a* (BChl *a*) chromophores whose excitonic couplings have been determined spectroscopically [21]. The site-energy Hamiltonian in the site basis (energies in cm^{-1} relative to $12\,210\,\text{cm}^{-1}$) is [21]:

$$H_{\text{FMO}} = \begin{pmatrix} 200 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 320 & 30.8 & 8.2 & 0.7 & 11.8 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 110 & -70.7 & -17.0 & -63.3 \\ 6.7 & 0.7 & -2.2 & -70.7 & 270 & 81.1 & -1.3 \\ -13.7 & 11.8 & -9.6 & -17.0 & 81.1 & 420 & 39.7 \\ -9.9 & 4.3 & 6.0 & -63.3 & -1.3 & 39.7 & 230 \end{pmatrix}. \tag{20}$$

The dominant energy transfer pathways proceed as $1 \to 2 \to 3$ and $6 \to 5/7 \to 4 \to 3$, with site 3 serving as the exit site coupled to the reaction center [21, 9]. Decoherence due to protein-solvent fluctuations occurs on timescales of $\sim 50\,\text{fs}$ at physiological temperature [9], while excitation transfer completes in ~ 1–$5\,\text{ps}$.

5.2 Dimensional collapse in FMO

For the seven-site FMO complex ($d = 7$):

- Full quantum state space: $d^2 - 1 = 48$ real dimensions
- Classical probability simplex: $d - 1 = 6$ dimensions
- Collapse ratio: $1/(d+1) = 1/8 = 12.5\%$

Decoherence annihilates $d^2 - d = 42$ off-diagonal (coherence) dimensions—seven-eighths of the information-geometric degrees of freedom. The surviving 6-dimensional simplex encodes the classical site populations $\{p_1, \ldots, p_7\}$ (with $\sum_k p_k = 1$), which describe the probability of finding the excitation at each chromophore.

Under Haken–Strobl dephasing [22] at rate γ and trapping at the reaction-centre site 3 at rate Γ_{trap}, the density matrix evolves as

$$\dot{\rho} = -\frac{i}{\hbar}[H_{\text{FMO}}, \rho] + \gamma \sum_{k=1}^{7} \left(|k\rangle\langle k|\rho|k\rangle\langle k| - \frac{1}{2}\{|k\rangle\langle k|, \rho\} \right) - \frac{\Gamma_{\text{trap}}}{2}\{|3\rangle\langle 3|, \rho\}. \tag{21}$$

The first term generates unitary evolution, the second dephases coherences at rate γ, and the third removes population into the reaction-centre sink ($\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$, following [7]). The transport efficiency is $\eta = \Gamma_{\text{trap}} \int_0^T \langle 3|\rho(t)|3\rangle\, dt$.

Since the trapping term makes the dynamics non-trace-preserving (population leaks irreversibly into the reaction centre), we define the *conditional state* $\hat{\rho}(t) = \rho(t)/\text{Tr}[\rho(t)]$—the state of the exciton conditioned on not yet having been trapped. This renormalized state is a proper density matrix to which the Bures metric and principal angle analysis apply. In the pure-dephasing limit ($H = 0$, $\Gamma_{\text{trap}} = 0$), off-diagonal elements decay as $\rho_{kl} \to e^{-\gamma t}\rho_{kl}$; the full dynamics (21) is a competition between coherent Hamiltonian evolution, incoherent dephasing, and trapping.

5.3 Transport efficiency and ENAQT

ENAQT [7, 6] occurs because pure quantum evolution can trap excitations in interference patterns, while moderate dephasing breaks destructive interference and opens transport

channels. Figure 2(a) shows the transport efficiency $\eta(\gamma)$ computed from (21) for both natural entry points: site 1 (from the baseplate) and site 6 (from the CsmA antenna).

Both initial conditions produce a clear ENAQT peak: efficiency rises from low values at $\gamma \to 0$ (quantum localization), reaches a maximum at $\gamma_{\text{opt}} \approx 100$–$150\,\text{cm}^{-1}$, and slowly declines at large γ (classical hopping limit). The optimal dephasing rate falls within the physiological range for FMO at 300 K ($\gamma \sim 50$–$200\,\text{cm}^{-1}$ [9]), consistent with the ENAQT literature [7].

5.4 Geometry of the quantum-classical split

We now ask: what does the ENAQT regime look like in the information geometry of the quantum-classical transition?

For the FMO complex ($d = 7$), the diagonal and off-diagonal tangent subspaces have dimensions $d - 1 = 6$ and $d^2 - d = 42$ respectively, and their relative orientation is described by $d - 1 = 6$ principal angles $\theta_1 \leq \cdots \leq \theta_6$. We compute these at each dephasing rate γ by evolving the full sink-inclusive model (21) for 5 ps, renormalizing the surviving state $\hat{\rho} = \rho/\text{Tr}[\rho]$, and extracting the Bures-metric cross-Gram matrix between diagonal and off-diagonal basis directions (see Appendix). Figure 2(b) shows the minimum principal angle $\theta_{\min}(\gamma)$ for both site 1 and site 6 initial excitations.

The two initial conditions produce nearly identical angle curves, confirming that the geometric structure of the quantum-classical transition is robust to the choice of entry point. At weak dephasing ($\gamma \lesssim 1\,\text{cm}^{-1}$), the minimum principal angle drops below 50°: the classical population subspace and the quantum coherence subspace are far from orthogonal, meaning that population and coherence parameters are strongly mixed in the Fisher information sense. At moderate dephasing ($\gamma \sim 5$–$20\,\text{cm}^{-1}$), angles are in the range 70°–85°. At the ENAQT optimal dephasing rate ($\gamma_{\text{opt}} \approx 100$–$150\,\text{cm}^{-1}$), all six angles exceed 87° and the state is geometrically close to classical. The dotted vertical lines in Figure 2(b) mark the efficiency-peak γ values from panel (a), showing that optimal transport occurs in a regime where the quantum-classical split is nearly orthogonal. Crucially, both panels are computed from the same sink-inclusive dynamics (21): the ef-

ficiency from accumulated trapped population, and the geometry from the renormalized conditional state $\hat{\rho}$.

This quantitative result reveals that ENAQT does not require a regime of strong quantum-classical mixing. The biologically relevant quantum effects arise from the *small residual departure* from classical orthogonality: the system exploits the last few degrees of geometric non-orthogonality before full classical decoupling. The most geometrically interesting transition—where angles pass through 45°–80°—occurs at dephasing rates ($\gamma \sim 1$–$20\,\mathrm{cm}^{-1}$) well below the physiological range.

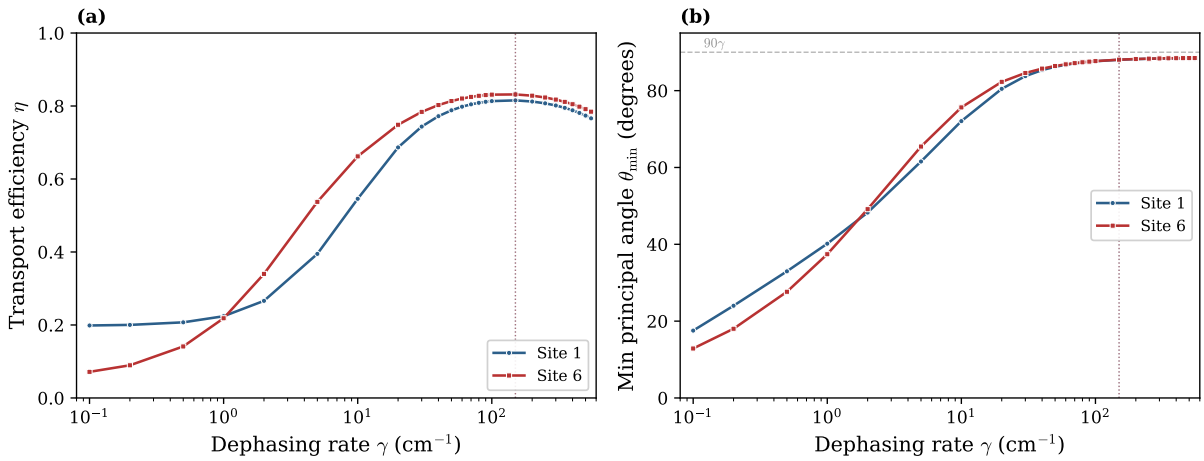

Figure 2: **Transport efficiency and Bures geometry in the FMO complex.** Both panels use the same sink-inclusive dynamics (21) with $\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$. **(a)** Transport efficiency η as a function of Haken–Strobl dephasing rate γ. Both natural entry points (site 1, site 6) show a clear ENAQT peak at $\gamma_{\mathrm{opt}} \approx 100$–$150\,\mathrm{cm}^{-1}$. **(b)** Minimum Bures principal angle $\theta_{\min}$ between the diagonal and off-diagonal tangent subspaces, computed from the renormalized conditional state $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$ at $t = 5\,\mathrm{ps}$. Dotted vertical lines mark the efficiency-peak γ from panel (a). At the ENAQT optimum, all principal angles exceed 87°: optimal transport occurs in a regime that is geometrically close to classical.

5.5 Comparison to scalar coherence measures

Standard coherence monotones—the ℓ_1-norm $\sum_{k\neq l} |\rho_{kl}|$ and relative entropy C_r—assign single scalars to the quantum-classical transition. They answer "how much coherence?" but not "how is coherence oriented relative to the classical subspace?" The $d-1$ principal angles provide a spectral decomposition of this geometric relationship. Figure 3(a) shows the full six-angle spectrum for FMO: at intermediate dephasing ($\gamma \sim 5$–$20\,\mathrm{cm}^{-1}$), the angles spread substantially, revealing that some subspace directions approach clas-

sical orthogonality faster than others. This heterogeneous structure—which directions decohere first and which persist longest—is invisible to any single scalar.

5.6 Robustness to trapping rate

The trapping rate $\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$ used above follows [7], but the actual rate depends on the FMO-reaction-centre coupling and varies across species. Figure 3(b) shows the minimum principal angle at the ENAQT-optimal dephasing rate as a function of Γ_{trap} across the range 0.2–5.0 ps^{-1}. For physiologically relevant trapping rates ($\Gamma_{\text{trap}} = 0.5$–$2.0\,\text{ps}^{-1}$), the minimum angle exceeds 86° for both initial conditions. Even at the aggressive rate $\Gamma_{\text{trap}} = 5\,\text{ps}^{-1}$, the angle remains above 81°—still well into the near-classical regime. The geometric conclusion is robust: faster trapping depletes population before dephasing fully completes, reducing the angle toward classical, but the qualitative picture holds.

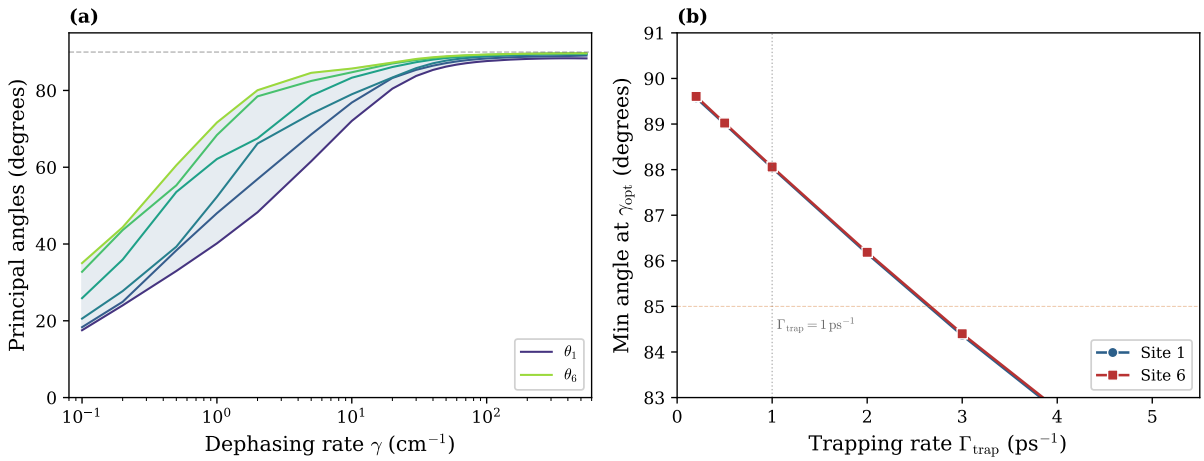

Figure 3: **Angle spectrum and model robustness. (a)** All six Bures principal angles $\theta_1 \leq \cdots \leq \theta_6$ between the diagonal and off-diagonal tangent subspaces as a function of dephasing rate γ (site-1 initial excitation, $\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$). The spread of angles at intermediate γ reveals heterogeneous geometric mixing across different subspace directions—structure invisible to scalar coherence measures. **(b)** Minimum principal angle at the ENAQT-optimal dephasing rate as a function of trapping rate Γ_{trap}. For physiologically relevant rates (0.5–2.0 ps^{-1}), angles exceed 86°; even at 5.0 ps^{-1} the system remains near-classical ($> 81°$).

5.7 Second complex: PE545

To test whether the geometric signature observed in FMO is specific to that complex or reflects a more general structure, we apply the same analysis to phycoerythrin 545

(PE545), a light-harvesting antenna from the cryptophyte alga *Rhodomonas* CS24 [23]. PE545 differs from FMO in several respects: it contains eight bilin chromophores (rather than seven bacteriochlorophylls), has approximate C_2 pseudo-symmetry, and uses phycoerythrobilin and dihydrobiliverdin pigments with different electronic properties. The excitonic Hamiltonian (Model E of [24]; see Appendix B) has site energies spanning $\sim 1\,600\,\mathrm{cm}^{-1}$—roughly four times the range in FMO—with maximum inter-site couplings of $92\,\mathrm{cm}^{-1}$.

For PE545, decoherence collapses the full quantum state space from $d^2 - 1 = 63$ dimensions to $d - 1 = 7$ classical dimensions (collapse ratio $1/9 \approx 11\%$). We model the irreversible step as Lindblad trapping at the two lowest-energy chromophores DBV19A and DBV19B (sites 1 and 2), which serve as the terminal energy acceptors in the antenna [24, 23]. The total trapping rate is $\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$, split equally between the two target sites. Initial excitation is placed at each of the two central-dimer sites PEB50/61C (site 0) and PEB50/61D (site 5), which absorb strongly and initiate energy transfer [25].

Figure 4(a) shows that PE545 exhibits a clear throughput-optimal dephasing regime: transport efficiency peaks at $\gamma_{\mathrm{opt}} \approx 700$–$750\,\mathrm{cm}^{-1}$ for both initial conditions. The optimum occurs at substantially higher dephasing than in FMO ($\gamma_{\mathrm{opt}} \approx 100$–$150\,\mathrm{cm}^{-1}$), consistent with the larger site-energy disorder requiring more noise to bridge energy gaps. Figure 4(b) shows the seven Bures principal angles as a function of dephasing. At γ_{opt}, all seven angles exceed 89°: the system is geometrically nearly classical at the transport optimum, reproducing the same qualitative signature observed in FMO. The conclusion is robust to the choice of target manifold (including PEB82C as a third target site shifts γ_{opt} by $< 50\,\mathrm{cm}^{-1}$ and changes minimum angles by $< 0.3°$) and to halving the trapping rate.

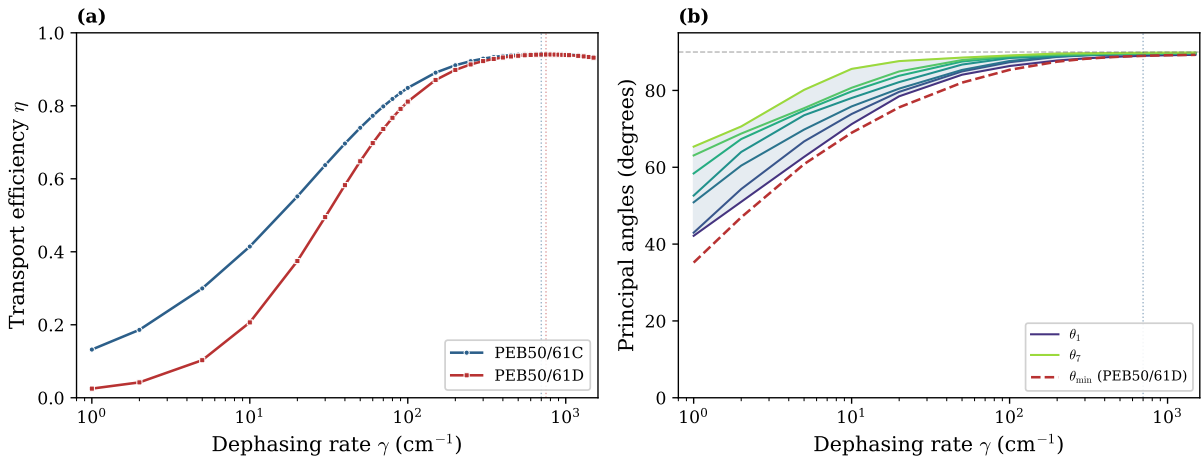

Figure 4: **Transport efficiency and Bures geometry in PE545.** Same analysis as Figure 2, applied to the eight-bilin PE545 complex ($d = 8$, collapse $63 \to 7$). Trapping at the two lowest-energy bilins (DBV19A, DBV19B) with $\Gamma_{\text{trap}} = 1\,\text{ps}^{-1}$. **(a)** Transport efficiency η as a function of dephasing rate γ for both central-dimer initial sites. The throughput-optimal regime peaks at $\gamma_{\text{opt}} \approx 700$–$750\,\text{cm}^{-1}$. **(b)** All seven principal angles for PEB50/61C initial excitation (solid, coloured by angle index), with the minimum principal angle for PEB50/61D overlaid (dashed red). The dotted vertical line marks γ_{opt} from panel (a). At the transport optimum, all angles exceed 89° for both initial conditions—the same near-classical geometric signature as in FMO.

The observation that two structurally distinct photosynthetic complexes—a bacteriochlorophyll trimer antenna (FMO, $d = 7$) and a bilin-based cryptophyte antenna (PE545, $d = 8$)—both exhibit near-classical Bures geometry at their respective transport optima suggests that this geometric property may be a generic feature of environment-assisted transport rather than an accident of FMO's particular Hamiltonian.

5.8 FMO coupling graph

The coupling structure of the FMO Hamiltonian can be represented as a graph where vertices are chromophore sites and edges connect pairs with significant electronic coupling ($|J_{kl}| > 5\,\text{cm}^{-1}$). Figure 5 shows this graph. Four site pairs have couplings below the $5\,\text{cm}^{-1}$ threshold: (2,5), (2,7), (3,5), and (5,7), all involving sites at opposite ends of the two main transport pathways $1 \to 2 \to 3$ and $6 \to 5/7 \to 4 \to 3$.

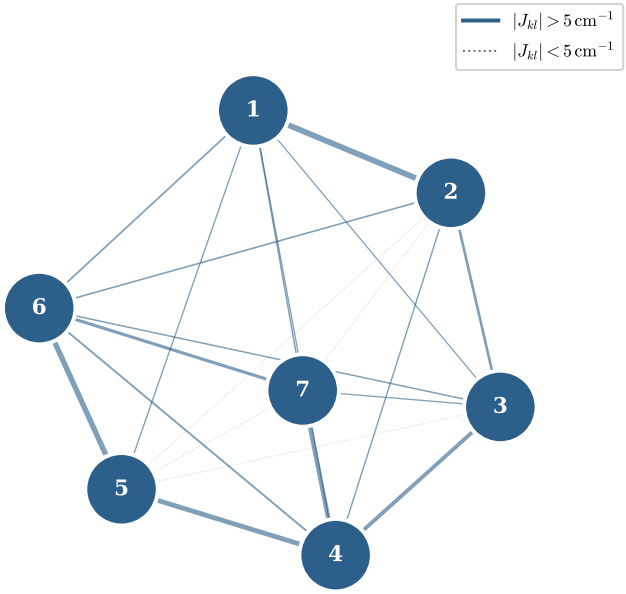

Figure 5: **FMO coupling graph.** Vertices represent the seven BChl sites; edges connect pairs with significant coupling ($|J_{kl}| > 5\,\mathrm{cm}^{-1}$). Edge widths are proportional to coupling strength. Dotted lines show weak couplings ($|J_{kl}| < 5\,\mathrm{cm}^{-1}$). The four weak pairs all involve sites at opposite ends of the two main transport pathways.

6 Application to Neural Oscillations

The principal angle framework developed in Sections 3–5 provides a continuous measure of "how quantum" any system is. Photosynthetic complexes sit near the transition (87°–89° at their respective ENAQT optima). Where do neural oscillator assemblies sit on this spectrum?

6.1 Neural oscillator assemblies

Working memory in primate cortex relies on high-dimensional mixed selectivity: individual neurons respond to conjunctions of task variables, and the population-level representation lives on a high-dimensional manifold maintained by gamma (30–70 Hz) and beta (15–30 Hz) oscillatory bursts [26, 27, 28, 29]. Pinotsis and Miller [30] have shown that stable electric field dynamics emerge from these oscillatory interactions, suggesting that the relevant dynamical degrees of freedom are the coupled oscillator populations of

cortical columns rather than individual neurons.

A cortical column can be modelled as $d \sim 10$–100 coupled oscillator populations spanning the gamma band [31]. The question is: where does such an assembly sit on the quantum-classical spectrum measured by the principal angle framework?

6.2 Dimensional collapse in a gamma-band microcircuit

We construct a $d = 10$ Hamiltonian representing ten coupled oscillator populations spanning the gamma band (30–70 Hz; see Appendix C for the full matrix). The coupling topology is a ring with two long-range shortcuts (small-world), matching the recurrent connectivity of cortical microcircuits. For $d = 10$:

- Full quantum state space: $d^2 - 1 = 99$ real dimensions
- Classical probability simplex: $d - 1 = 9$ dimensions
- Collapse ratio: $1/(d + 1) = 1/11 \approx 9\%$

The Lindblad framework is identical to the photosynthetic analysis: Haken–Strobl dephasing at rate γ with trapping at two target sites ($\kappa = 0.5\,\mathrm{ps}^{-1}$ each in scaled units). The crucial property is *scale invariance*: the Lindblad equation under $(H, \gamma, \kappa, t) \to (\alpha H, \alpha\gamma, \alpha\kappa, t/\alpha)$ preserves $\rho(t)$ and therefore all principal angles. Results depend only on the dimensionless ratio $\gamma/J_{\max}$, not absolute energy. We scale the neural Hamiltonian from Hz to cm^{-1} ($\alpha = 50\,\mathrm{cm}^{-1}/\mathrm{Hz}$) so that matrix entries are $O(10^3)$, matching the FMO/PE545 range.

6.3 The detection threshold

At conservative thermal dephasing ($T = 310\,\mathrm{K}$), the biological dephasing rate for neural oscillations is $\gamma_{\mathrm{bio}} = k_B T/\hbar \approx 4 \times 10^{13}\,\mathrm{Hz}$, giving $\gamma_{\mathrm{bio}}/J_{\max} \approx 8 \times 10^{12}$. At this ratio, all nine principal angles are $90.00°$ to machine precision: the system is geometrically indistinguishable from the classical simplex.

We define a *detection threshold* at $\theta_{\min} = 85°$: below this angle, quantum-classical mixing is at least as strong as FMO at its biological operating point. The detection

threshold occurs at $\gamma/J_{\max} \sim O(1)$, the same dimensionless ratio as in photosynthetic systems. To reach this regime from biological dephasing, the effective dephasing rate would need to be reduced by ∼12 orders of magnitude from bulk thermal estimates (Figure 6b). For comparison, photosynthetic protein baths provide ∼2–3 orders of magnitude of shielding from bulk thermal dephasing [9]—insufficient by ∼9 orders for neural oscillation frequencies.

This does not rule out quantum effects in shielded microenvironments (ion channel pores, protein interiors, microtubule lumens) that operate at molecular energy scales rather than oscillation frequencies. The framework is agnostic: it quantifies the detection threshold for any claimed quantum effect as the dephasing reduction needed to bring principal angles below 90°. Tegmark [32] reached a similar conclusion via decoherence timescale estimates; ours is a complementary geometry argument that provides a continuous bound rather than a binary timescale comparison.

6.4 The geometric spectrum across biological scales

Figure 6 places the three systems on a single axis of dimensionless dephasing $\gamma/J_{\max}$:

- **FMO** ($d = 7$): $\gamma_{\text{bio}}/J_{\max} \approx 1.1$, near the ENAQT optimum. Principal angles 87°–90°. Quantum effects are geometrically exploitable.
- **PE545** ($d = 8$): $\gamma_{\text{bio}}/J_{\max} \approx 7.9$, slightly past the optimum. Angles $> 89°$. Still in the transition zone.
- **Neural gamma** ($d = 10$): $\gamma_{\text{bio}}/J_{\max} \approx 8 \times 10^{12}$. Angles pinned at 90.00°. Deep classical.

These three systems span 12 orders of magnitude in $\gamma_{\text{bio}}/J_{\max}$, yet all exhibit the same ENAQT-equivalent geometry at $\gamma/J_{\max} \sim O(1)$, confirming the scale invariance of the principal angle framework.

The result connects to the classical coherence program: neural computation operates *within* the Fisher–Rao simplex, exploiting classical coherence—oscillatory phase relationships, high-dimensional population dynamics—rather than quantum coherence [33, 34].

The principal angle framework converts the qualitative claim "neural systems are classical" into a quantitative geometric bound: the minimum microenvironment shielding factor needed for any quantum effect to become geometrically distinguishable.

As an empirical validation, we repeat the analysis with the Potjans–Diesmann canonical cortical microcircuit model ($d = 8$, empirical connectivity from [35]; see Appendix C). Both the synthetic ($d = 10$) and empirical ($d = 8$) neural models give the same qualitative result: all angles $\approx 90°$ at biological dephasing, with the ENAQT-equivalent regime at $\gamma/J_{\max} \sim O(1)$.

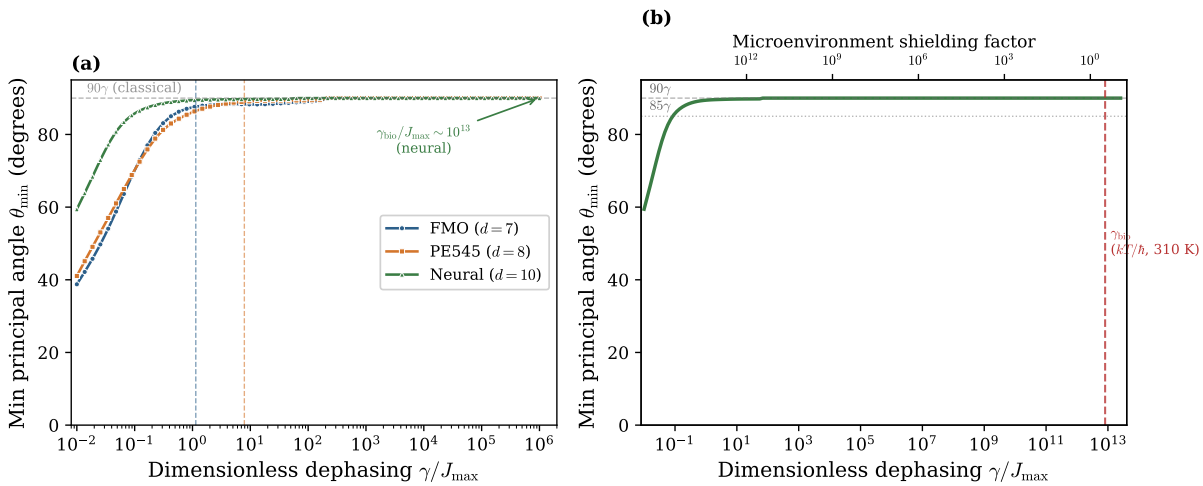

Figure 6: **Cross-scale quantum-classical boundary. (a)** Minimum Bures principal angle $\theta_{\min}$ as a function of dimensionless dephasing $\gamma/J_{\max}$ for FMO ($d = 7$, blue), PE545 ($d = 8$, orange), and a cortical gamma-oscillation assembly ($d = 10$, green). Dashed vertical lines mark biological dephasing for FMO and PE545; the neural biological dephasing ($\gamma_{\text{bio}}/J_{\max} \sim 10^{13}$) lies far off-scale to the right. All three systems exhibit the same transition geometry at $\gamma/J_{\max} \sim O(1)$, confirming scale invariance. **(b)** Detection threshold for the neural system: $\theta_{\min}$ vs. hypothetical dephasing, with the biological dephasing marked (dashed red). The top axis shows the microenvironment shielding factor (ratio of biological to effective dephasing) that would be needed to enter the geometrically interesting regime. Approximately 12 orders of magnitude of shielding are required—far beyond what photosynthetic protein baths provide (∼2–3 orders).

7 Discussion

7.1 Summary of results

We have shown that decoherence is a dimensional collapse in the information-geometric sense: $d^2 - 1 \to d - 1$ dimensions, with the Bures metric restricting to Fisher–Rao on

the surviving simplex. The exact Bures separation angle quantifies the geometry of the classical-quantum split at every full-rank qubit state, and the global orthogonality theorem (Theorem 3.8) establishes that for $d \geq 3$ the diagonal and off-diagonal tangent subspaces are Bures-orthogonal *only* at diagonal states—any coherence breaks orthogonality, resolving the higher-dimensional geometry exactly. Applied to three biological systems spanning 12 orders of magnitude in dephasing—FMO ($d = 7$), PE545 ($d = 8$), and a cortical gamma assembly ($d = 10$)—the principal angle framework reveals a continuous geometric spectrum: photosynthetic complexes sit near the quantum-classical boundary (angles 87°–89° at their ENAQT optima), while neural oscillator assemblies are geometrically indistinguishable from the classical simplex at oscillation-frequency scales (90.00° to machine precision). The framework converts "is it quantum?" from a binary question into a quantitative geometric bound.

7.2 Limitations

The qubit separation angle (Theorem 3.5) is exact, and the global orthogonality theorem (Theorem 3.8) is sharp for all $d \geq 3$; the perturbative expansion (Theorem 3.10) quantifies the departure from orthogonality but is limited to small coherence. The Haken–Strobl dephasing model used for FMO is simplified: the real protein-solvent bath has structured spectral density, non-Markovian dynamics, and spatially correlated fluctuations [9]. The principal angles are computed on the renormalized conditional state $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$; this is the natural object for the surviving exciton's geometry, but it means the Bures analysis inherits any artifacts of the Haken–Strobl model's treatment of system-bath correlations. Extending the information-geometric analysis to structured baths and non-Markovian dephasing is an open problem.

7.3 Outlook

Theorem 3.3 establishes that the Fisher–Rao metric on the surviving classical simplex is the restriction of the Bures metric from the quantum state manifold. The neural application (Section 6) demonstrates that this geometric continuity holds across 12 orders

of magnitude in dephasing: the same framework that quantifies quantum-classical mixing in photosynthesis also characterizes the detection threshold for quantum effects in neural systems. Classical coherence phenomena on longer timescales [33, 34]—dissipative dynamics within the Fisher–Rao simplex—can in principle be analyzed with the same information-geometric tools developed here, now that the boundary between quantum and classical regimes has been quantified.

Extending the framework to molecular-scale neural microenvironments (ion channels, synaptic receptors, protein interiors) where energy scales approach the photosynthetic regime remains open. Whether quantum effects in such microenvironments could propagate to oscillation-level dynamics through nonlinear amplification is a separate question that the principal angle framework could address given appropriate Hamiltonians.

7.4 Open questions

1. *Non-Markovian geometry*: characterizing the dimensional collapse under structured (non-Markovian) dephasing, where coherence revivals temporarily restore collapsed dimensions.

2. *Multi-chromophore entanglement*: extending beyond single-exciton states to multi-exciton manifolds, where inter-chromophore entanglement adds structure beyond site-basis coherence.

3. *Quantum Darwinism*: the pointer states selected by einselection [3, 36] determine *which* $(d-1)$-dimensional submanifold the collapse targets; the collapse magnitude is basis-independent.

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A Numerical Methods

Random state generation. Full-rank density matrices are drawn from the Hilbert–Schmidt measure: generate a $d \times d$ matrix A with i.i.d. standard complex Gaussian entries, form $\rho = AA^\dagger / \mathrm{Tr}[AA^\dagger]$.

SLD computation. For each tangent direction X, the SLD L_X satisfying $\rho L_X + L_X \rho = 2X$ is computed in the eigenbasis: $(L_X)^{\mathrm{eig}}_{mn} = 2X^{\mathrm{eig}}_{mn}/(\lambda_m + \lambda_n)$.

Principal angles. The $d-1$ principal angles between T_ρ^{diag} and T_ρ^{off} are computed as $\theta_a = \arccos(\sigma_a)$, where $\sigma_1, \ldots, \sigma_{d-1}$ are the singular values of $G_{DD}^{-1/2} G_{DO} G_{OO}^{-1/2}$.

Adversarial search. 100 L-BFGS-B optimizations per dimension maximize $\min_a \theta_a$ over full-rank parameterized states. No optimized state approaches 90°.

FMO dephasing simulation. The Lindblad master equation (21) is integrated numerically using a fourth-order Runge–Kutta scheme with time step $\Delta t = 1.0\,\mathrm{fs}$. The FMO Hamiltonian (20) uses site energies and couplings from Adolphs and Renger [21]. Both transport efficiency and principal angles are computed from the same sink-inclusive dynamics ($\Gamma_{\mathrm{trap}} = 1\,\mathrm{ps}^{-1}$ at site 3). For the efficiency, accumulated trapped population is integrated over $t = 15\,\mathrm{ps}$. For the principal angles, the density matrix is evolved for $t = 5\,\mathrm{ps}$ and the surviving state is renormalized: $\hat{\rho} = \rho/\mathrm{Tr}[\rho]$. Both initial conditions $\rho(0) = |1\rangle\langle 1|$ (site 1) and $\rho(0) = |6\rangle\langle 6|$ (site 6) are used to test robustness. For dephasing rates $\gamma \geq 5\,\mathrm{cm}^{-1}$, the coherence decay time $\gamma^{-1} \lesssim 1\,\mathrm{ps}$ is shorter than the evolution time, ensuring coherences have reached a quasi-stationary balance. At physiological dephasing rates ($\gamma \geq 50\,\mathrm{cm}^{-1}$), the minimum principal angle varies by less than 1.2° across readout times of 3–8 ps, confirming that the geometric conclusions are not sensitive to the choice of evolution window. The renormalized state is regularized to full rank ($\hat{\rho} \mapsto \hat{\rho} + 10^{-10}\, I/d$) before computing the six Bures principal angles between the diagonal and off-diagonal tangent subspaces.

Software. All computations use NumPy and SciPy. Source code is available in the repository.

B PE545 Hamiltonian and Parameters

The PE545 Hamiltonian uses site energies and couplings from Model E of Novoderezhkin et al. [24]. Site ordering: $0 = \mathrm{PEB50/61C}$, $1 = \mathrm{DBV19A}$, $2 = \mathrm{DBV19B}$, $3 = \mathrm{PEB82C}$,

4 = PEB158C, 5 = PEB50/61D, 6 = PEB82D, 7 = PEB158D.

$$H_{\mathrm{PE545}} = \begin{pmatrix} 18532 & 1 & -37 & 37 & 23 & 92 & -16 & 12 \\ 1 & 18008 & 4 & -11 & 33 & -39 & -46 & 3 \\ -37 & 4 & 17973 & 45 & 3 & 2 & -11 & 34 \\ 37 & -11 & 45 & 18040 & -7 & -17 & -3 & 6 \\ 23 & 33 & 3 & -7 & 18711 & 18 & 7 & 6 \\ 92 & -39 & 2 & -17 & 18 & 19574 & 40 & 26 \\ -16 & -46 & -11 & -3 & 7 & 40 & 19050 & 7 \\ 12 & 3 & 34 & 6 & 6 & 26 & 7 & 18960 \end{pmatrix} \mathrm{cm}^{-1}. \quad (22)$$

The diagonal entries are site energies; off-diagonal entries are electronic couplings. Sites 1 and 2 (DBV19A, DBV19B) are the lowest-energy bilins forming the "red pool" that serves as the terminal energy acceptor. Initial excitation is placed at sites 0 or 5 (the central PEB50/61 dimer). The Lindblad trapping sink is applied at the target manifold $\{|1\rangle, |2\rangle\}$ with $\kappa_1 = \kappa_2 = 0.5\,\mathrm{ps}^{-1}$ ($\Gamma_{\mathrm{trap}} = 1.0\,\mathrm{ps}^{-1}$ total). All other model parameters (dephasing model, RK4 integrator, regularization) are identical to the FMO analysis (Appendix A).

Table 1 reports the robustness of PE545 results to target-manifold definition and trapping rate, backing the claims in Section 5.7.

Table 1: **PE545 robustness check.** Transport optimum γ_{opt}, peak efficiency η, and minimum Bures principal angle θ_{min} at γ_{opt} (readout at $t = 5\,\mathrm{ps}$) for three model variants. "Default": target sites $\{1, 2\}$, $\Gamma_{\mathrm{trap}} = 1.0\,\mathrm{ps}^{-1}$. "Alt. target": target sites $\{1, 2, 3\}$, same total rate. "Half κ": target sites $\{1, 2\}$, $\Gamma_{\mathrm{trap}} = 0.5\,\mathrm{ps}^{-1}$.

Condition	Initial site	γ_{opt} (cm^{-1})	η	θ_{min}
Default	PEB50/61C	725	0.943	89.0°
	PEB50/61D	775	0.941	89.1°
Alt. target	PEB50/61C	725	0.950	89.2°
	PEB50/61D	775	0.947	89.2°
Half κ	PEB50/61C	700	0.800	89.5°
	PEB50/61D	800	0.797	89.5°

C Neural Microcircuit Parameters

Primary model: gamma-band cortical microcircuit ($d = 10$). Ten coupled oscillator populations spanning the gamma band (30–70 Hz), modelling a cortical column. Site labels: L4in (layer-4 input), L4a, L4b, L2/3a, L2/3b, L2/3c, L5a, L5b, L6a, L6b. Coupling topology: ring with nearest-neighbour coupling ($J = 5\,\mathrm{Hz}$) plus two long-range shortcuts (sites $1 \leftrightarrow 6$ and $3 \leftrightarrow 8$, $J = 2\,\mathrm{Hz}$), producing small-world connectivity consistent with cortical column architecture [31].

Site energies (Hz): $\{30, 33, 36, 40, 44, 48, 53, 58, 64, 70\}$.

The Hamiltonian is scaled to cm^{-1} by $\alpha = 50\,\mathrm{cm}^{-1}/\mathrm{Hz}$ for numerical compatibility with the Lindblad integrator. By scale invariance of the Lindblad equation (Section 6.2), all principal angles depend only on the dimensionless ratio $\gamma/J_{\max}$.

Physical parameters:

- Site energy spread: $\Delta E \approx 40\,\mathrm{Hz}$
- Maximum coupling: $J_{\max} = 5\,\mathrm{Hz}$ (nearest-neighbour)
- Biological dephasing (thermal): $\gamma_{\mathrm{bio}} = k_B T/\hbar \approx 4 \times 10^{13}\,\mathrm{Hz}$ at $310\,\mathrm{K}$
- Dimensionless ratio: $\gamma_{\mathrm{bio}}/J_{\max} \approx 8 \times 10^{12}$

Initial excitation: site 0 (L4 input), analogous to thalamic drive. Target manifold: sites 8, 9 (lowest-frequency populations—energy flows downhill in frequency). Trapping rate: $\kappa = 0.5\,\mathrm{ps}^{-1}$ per target site (in scaled units).

Empirical validation: Potjans–Diesmann cortical column ($d = 8$). The canonical cortical microcircuit model of Potjans and Diesmann [35] provides a published $d = 8$ connectivity matrix: four layers $\times$ {excitatory, inhibitory} = L2/3e, L2/3i, L4e, L4i, L5e, L5i, L6e, L6i. Connection probabilities and relative synaptic weights are from Table 5 of [35]. We symmetrize the connection probability matrix via $J_{ij}^{\mathrm{eff}} = \sqrt{C_{ij}\,C_{ji}}$ and scale to the cm^{-1} range. Initial excitation at L4e (thalamic input); target manifold L5e, L6e (cortical output layers).

Table 2: **Cross-scale comparison.** Optimal dimensionless dephasing $\gamma_{\text{opt}}/J_{\text{max}}$, minimum principal angle at the transport optimum $\theta_{\text{min}}^{\text{opt}}$, and minimum principal angle at biological dephasing $\theta_{\text{min}}^{\text{bio}}$ for all four systems. At biological dephasing, both neural models give $\theta_{\text{min}} = 90.00°$ to machine precision.

System	d	$d^2{-}1 \to d{-}1$	$\gamma_{\text{opt}}/J_{\text{max}}$	$\theta_{\text{min}}^{\text{opt}}$	$\theta_{\text{min}}^{\text{bio}}$
FMO	7	$48 \to 6$	$\sim$1–2	87°–89°	87°–88°
PE545	8	$63 \to 7$	~ 8	89°	89°
Neural (synthetic)	10	$99 \to 9$	$\sim O(1)$	89.8°	90.00°
Neural (PD empirical)	8	$63 \to 7$	$\sim O(1)$	89.5°	90.00°

References for neural parameters: gamma oscillation mechanisms and cortical column architecture [31]; gamma/beta bursts in working memory [26]; working memory 2.0 [29]; high-dimensional mixed selectivity [28, 27]; cytoelectric field dynamics [30]; canonical cortical microcircuit [35]; decoherence timescale estimates for neural systems [32].